

MEMORANDUM

RHS Cornerstone Limited

Date: Tuesday, February 3rd, 2026

Subject: Request for Proposals for a Search-and-Rescue Mission Catapult Design & Scale Prototype

The purpose of this memo is to inform all interested design consulting firms of a request to submit proposals for the **design and scale functional prototype** of a **search-and-rescue mission catapult** for our company.

Summary

Rashidzadeh, Heibati, & Salfi (RHS) Cornerstone Limited is seeking design proposals for an automated mechanical catapult-based search-and-rescue delivery system.

The final design proposal will include a written proposal, poster presentation, and functional scale prototype. The proposed system must be electrically powered and be physically represented at a reduced scale using an Arduino ELEGOO UNO R3 Project Smart Robot Car Kit V4 as the base for the functional prototype. The Robot Car will serve to autonomously navigate a predesigned course and stop at a designated rescue drop zone, while a mounted or tow-behind catapult is to be designed for the efficient, safe, timely, and consistent distribution of three (3) emergency supply payloads launched consecutively at the designated area. The catapult must be physically represented by a constructed scale prototype using materials selected and acquired by your design team.

An **aesthetic representation** of the catapult-based search-and-rescue delivery system design concept must be represented on the prototype as constrained by the Arduino ELEGOO UNO R3 Project Smart Robot Car Kit V4 and relative scale. The catapult component (with all bases and attachments) must not exceed the size of the robot car in width or length (there is no restriction on height). The sketch and physical representation will be competitively evaluated in our selection of the final design for implementation.

The functional prototype must be programmed to traverse a preset simulation course that will be revealed to the design groups at a later date. Programming elements that may be required to complete the simulation course include, but are not limited to, line-following and object avoidance. The prototype vehicle must operate fully autonomously (i.e., no use of remote / app-based driving function).

The prototype catapult must be connected to the robot car in a manner that is non-destructive (i.e., no permanent modification to the robot kit is permitted). The launcher must be a mechanical device that uses stored energy (spring, elastic element, torsion mechanism, lever, or equivalent). Alternative designs are allowed provided they are safe, consistent, and do not interfere with the robot's sensors, wheels, or stability.

Your custom designed and manufactured catapult must be designed for the safe, accurate, and effective distribution of emergency supply payloads in search-and-rescue missions. It will be important for your engineering design team to understand relevant standards, codes, and regulations for safety, performance, and machine design, and other pertinent considerations.

During the evaluation phase, each proposing design team will be provided with three (3) emergency supply payload units which will represent the emergency supply payload to be distributed by the search-and-rescue

catapult system. Figure 1 provides an example of simulated emergency supply payload units: ping pong balls. Your prototype should be designed to carry and distribute **all** provided emergency supply payload units. A key functional element that will be used in evaluating the designs is the successful completion of the course with the robot car and attached catapult. The scale prototype mechanical catapult-based rescue delivery system design must also be represented in a fully detailed set of AutoCAD drawings.



Figure 1: Simulated Payload Units

Design teams are responsible for the following:

- a) Designing and fabricating the launcher and autoloader.
- b) Writing the Arduino program to control autonomous navigation, stopping, automatic reloading, and launching operations.

Key functional elements that will be evaluated include

- a) fully autonomous operation (no remote control);
 - b) mechanical launcher using stored energy;
 - c) mandatory autoloader (must reload the launcher automatically between launches);
 - d) complete stop at the rescue drop zone before launch;
 - e) consecutive launch of exactly three standard ping pong balls;
 - f) target accuracy (only balls landing fully inside the target bucket count as successful delivery);
- and,
- g) no robot interference (launcher and autoloader must not interfere with robot stability, sensors, or wheels).

Important Notes:

- A custom designed and constructed catapult system must be developed, and the use of any existing commercially scale-sized catapults (e.g., scale models, toys, etc.) is not permissible. Conversely, your design team is permitted to utilize scale-size commercially available parts (e.g., toy wheels).
- The Arduino ELEGOO UNO R3 Project Smart Robot Car Kit V4 must not be damaged or destructively modified. The connection of the catapult system to the robot car must be non-destructive (i.e., no drilling, sawing, etc. that would permanently modify the robotic kit).

Design Proposals

All written design proposals will follow the provided template and must include a definition of the design problem and requirements; background research (including appropriate references in APA 7 format) to help establish all elements of the proposed solution; a thorough discussion of the proposed design (including design sketch(es)); a discussion and table breakdown of the cost of the proposed scale prototype design; a conclusive summary; and an implementation recommendation. Additionally, the submission must include the following accompanying appendices:

- an instruction set for the construction/fabrication procedure of the search-and-rescue delivery system; and,
- AutoCAD generated drawing(s) of the proposed catapult system designs specifically used and represented on the scale prototype vehicle.

Additional Deliverables

It is important to RHS Cornerstone Limited to understand and monitor the design process for this RFP, and as such, the following progress activities and associated documents will be evaluated throughout your design effort:

- an ideation report including at least one proposed design solution (including description and AutoCAD sketch/illustration) from each design team member and a decision matrix that exhibits the team's collective decision process for the proposed solution;
- a concept proposal and progress status report following a provided template that will briefly pitch the design and exhibit the team's project plan and timeline, member responsibilities, and project status;
- a video recording of a 2-minute pitch presented by all design team members presenting the initial concept, providing context of how it solves the design problem as the team defined it;
- mandatory attendance of two progress check-in meetings with the team's assigned project director (i.e., graduate assistant or teaching assistant) and submission of associated meeting agendas and minutes; and,
- full programming code utilized for the Arduino ELEGOO UNO R3 Project Smart Robot Car Kit V4.

Twice in the duration of the project, midway through and nearing completion, all team members will be required to complete a self and peer evaluation of their team's performance.

Finally, all design groups must prepare and submit a Poster Presentation that will be presented, in person, by ALL team members on March 31st, 2026, along with their constructed prototype.

Required Project Documents & Artifacts

The following documentation and artifacts will be required from all design teams:

1. Each team will submit one proposal. The proposal must be formatted according to the provided template. The proposal should not exceed (8) pages in length (excluding title page, references, and appendices). The proposal shall include at least one embedded illustration (sketch or drawing) of the proposed design.
2. Each team will create one functional scale prototype. The prototype will be created using an Arduino ELEGOO UNO R3 Project Smart Robot Car Kit V4 for the vehicle base. The prototype must be fully functional to meet the requirements of the competitive simulation course and must provide an aesthetic representation of the design concept. The mechanical catapult prototype must be constructed to launch emergency supply payloads to a designated area. The prototypes will be evaluated on the simulation course on March 18th (trial 1) and 23rd (trial 2), during the regular meeting periods. A prototype Performance Competition will occur on March 24th, 2026.
3. Each team will prepare a Poster Presentation board. The board size will be 24-in x 36-in, and it will be physically available for evaluation (poster presentation) on March 31st, 2026. A quality image of the board must be submitted as PDF via Brightspace as well. The in-person poster presentation, also delivered on March 31st, 2026, should not exceed 3 minutes.
4. Each team will submit the full programming code. The code must be the code utilized for the Arduino ELEGOO UNO R3 Project Smart Robot Car Kit V4 during the prototype competitive simulation.
5. Each team will submit a set of engineering drawing(s). The drawing(s) must provide an accurate representation of the scale prototype representation of the mechanical catapult-based rescue delivery system. The drawing must be prepared in AutoCAD both in DWG and PDF format, appropriately laid out and detailed according to engineering practices acquired in GENG-1102 – Engineering Graphics.
6. Each team will submit an ideation report and decision matrix. The report will define the problem and requirements (i.e., functions, objectives, and constraints) and discuss how they have been prioritized. The report will discuss and illustrate (i.e., AutoCAD sketch/drawing) at least one design option from each team member's brainstorming activity. A decision matrix will be provided including a conclusion rationalizing the design selected to move forward.
7. Each team will submit a concept proposal and status report. A template will be provided, and the report will require a brief pitch of the proposed concept and a project plan inclusive of a Gantt chart, the project status as it relates to the timeline, and division of team member tasks and contributions to date.
8. Each team will submit a concept proposal pitch video recording. All team members will share in the presentation delivery of a 120-second video recording pitching the proposed rescue delivery system design and how it will best solve the problem as the team has defined it. Peer evaluation of the concept proposal pitch video recordings will form a portion of the team grade. Teams must participate in peer evaluation to earn a peer evaluation assessment as part of the rubric.
9. Each team will attend and participate in (2) progress check-in meetings. These meetings will be led by the team's assigned project director (i.e., graduate or teaching assistant) and require full attendance. The teams will be required to provide an agenda and take minutes for the meeting and later submit this

documentation (i.e., agenda and minutes) on Brightspace. The meetings will occur during the assigned meeting times on Mondays and Wednesdays (specific deadlines discussed in the following section).

10. Each team member will be required to complete a midway and final survey-based self/peer member evaluation. The surveys will be identical and will focus on team member contribution and performance and provide an opportunity to provide professional feedback.

Project Deadlines

1. All final proposal documents and final self/peer evaluation survey (items 1, 5, and 10) must be submitted **electronically by 11:59 PM on Monday, April 6th**.
2. All Poster Presentations (item 3) will be reviewed by our Engineering Management Team and the University of Windsor Faculty of Engineering on **Tuesday, March 31st**, from 8:30 AM -11:30 AM. Casual business attire is required. Attendance and participation of the entire design team required.
3. The functional prototype (**item 2**) must be available for display on **Tuesday, March 31st** for the Poster Presentation and be ready and available for operation during the Prototype Performance Evaluations on **Wednesday, March 18th and Monday, March 23rd**. If selected, it must also be available for the Prototype Performance Competition on **Tuesday, March 24th**. Competition course details will be disclosed in March.
4. The PDF image of the Poster Presentation board (**item 3**) is due in Brightspace on **Tuesday, March 31st at 11:59 PM**.
5. A Word document with all programming code (**item 4**) is due on the day of Prototype Performance Evaluations, Trial #2 (**March 23rd**) at 11:59 pm in Brightspace
6. The Ideation Report and Decision Matrix (**item 6**) is due in Brightspace on **Friday, February 27th, 2026 at 11:59 PM**.
7. The Midterm Peer Evaluation will be conducted via a web survey which will start on **Tuesday, February 24th** and must be completed by **Monday, March 2nd at 11:59 PM**.
8. The Concept Proposal & Status Report and Concept Proposal Pitch recording (**items 7 and 8**) are due in Brightspace on **Sunday, March 8th @ 11:59 PM**. Concept Proposal Pitch recordings will be peer-reviewed during the lecture period on Tuesday, March 10th. This peer review will form a portion of the team assessment. Participation in the peer review is **mandatory**.
9. Progress check-in meetings (**item 9**) will be completed during the scheduled tutorial/lab time for your section. Your team will meet with the Project Director (i.e., graduate or teaching assistant) as follows:

Requirement	Deadline
Progress Check-in Meeting #1	Wednesday, February 25 th , during your regularly scheduled lab/tutorial time.
Progress Check-in Meeting #2	Wednesday March 11 th , during your regularly scheduled lab/tutorial time.

Evaluation Structure

Part 1: Mechanical & Design Evaluation (Out of 100 points)

Part 1 evaluates engineering quality, robustness, and reliability and contributes **5%** toward the total course grade.

Component	Points	Description
Autoloader Functionality	50	Reliable reloading sequence, no jams, fully autonomous
Mechanical Robustness	25	Secure mounting, repeatable launches, structural stability, does not interfere with sensors
Design Quality, Innovation, and RFP Compliance	25	Simplicity, aesthetics, creative engineering solutions, RFP compliance

Part 2: Delivery / Competition Ranking (Open Ranking)

Part 2 rewards fast, accurate, and autonomous performance, while penalizing interventions and contributes **5%** toward the total course grade.

Competition Score

Participants will be evaluated based on mission performance using the following formula:

$$\text{Competition Score} = \text{Total Time (seconds)} - 10 \times (\text{Number of Balls Landed}) + 5 \times (\text{Number of Interventions})$$

- Total Time: Time in seconds from the start signal to the completion of the third launch
- Number of Balls Landed: Number of ping pong balls landing fully inside the bucket (max 3)
- Number of Interventions: Number of times a human intervened to assist the robot

Ranking rules:

- The lowest competition score wins.
- To scale scores for grading, the best performance is 100 points, the worst is 0 points.

Intermediate scores are scaled linearly:

$$\text{Scaled Score} = 100 \times (\text{Worst Score} - \text{Team Score}) / (\text{Worst Score} - \text{Best Score})$$

Example:

Worst total score = 20, best total score = 14, a team's score = 16 → Scaled Score = $100 \times (20-16)/(20-14)$
= 66.7

Note: Together, Part 1 and Part 2 account for **10% of the total course grade**, as outlined in the course syllabus under Prototype Performance.